☐ EXE Document: Issued for Review

☐ EXE Document: Issued for Construction

A	13/01/25	ADE	ACO	AHE	First issue
REV	**DATE**	**AUTHOR**	**AUDITOR**	**VALIDATOR**	**MODIFICATION**

Biscay Gulf Project

Electrical Converter Stations

EMPLOYER:

INELFE

GESTIONNAIRE DU RESEAU DE TRANSPORT D'ELECTRICITE
GMR GASCOGNE
12, rue Aristide Bergès, 33270 FLOIRAC Tel : 05 56 33 99 01

CONSORTIUM:

HITACHI ENERGY SWEDEN AB
HITACHI ENERGY FRANCE
HITACHI ENERGY SPAIN S.A.U.
VINCI-ENERGIES TRANSPORT ET TRANSFORMATION D'ENERGIES
CHANTIERS MODERNES NOUVELLE AQUITAINE

Architect :
Architecture MALACAN MARTRES
Immeuble Buzzichelli, 63 boulevard Carnot
31 000 Toulouse

LOGO ENTREPRISE EXE

Detailed Design:
Entreprise
Adresse Entreprise

CODE WILLYDOC (if Necessary): CODE WILLYDOC

Medium Voltage Switchgear (MVS) - Requirement Specification

Scale: N/A

Size: A4

DOCUMENT NUMBER:

BGP-CS-EM-F-PEL-TSP-0030

DISCIPLINE:

PEL

VINCI-Energies Transport et Transformation d'Energies

14 Bis impasse René Couzinet
31500 TOULOUSE
France métropolitaine

MODIFICATIONS LIST

REVISION	***DATE***	***MODIFICATIONS***
A	13/01/2025	First issue

SUMMARY

1 GENERAL

The medium voltage switchgears will be used for supply of power to the station service distribution transformers in Station 2 (Cubnezais) Converter Stations.

This specification is generated as a document that describes the minimum requirements for the medium voltage switchgear at the Cubnezais converter station.

Customer requirements and local standards will be mandatory. The strictest requirements must be followed.

The specification discribs the French RTE power supply (NF C 13-200) and the ENEDIS power supply (NF C 13-100).

1.1 Terminology

The abbreviations used in this document are following:

CT	Current Transformer
FO	Fibre Optical
HW	Hard Wired
IED	Intelligent Electronic Device
ITP	Inspection and Test Plan
MCB	Miniature Circuit Breaker
MVS	Medium Voltage Switchgear
LVS	Low Voltage Switchgear
HMI	Human machine interface
TCP	Transmission Control Protocol
FAT	Factory Acceptance Test
TTSR	Type Test Summary Report
MACH	ABB Control system
AIS	Air Insulated Switchgear
NO/NC	Normally Open/Normally Closed

1.2 Standards

The Medium voltage switchgear shall be designed, rated and tested in accordance with NFC 13-200 and other IEC, NF EN standards relevant to apparatus included in the switchgear.

NF EN 62271-1	High voltage switchgear and control gear – common specifications, including amendment 1 (2011).
NF EN 62271-100	High voltage switchgear and control gear – Alternating circuit breakers
NF EN 62271-102	High voltage switchgear and control gear – Alternating current disconnectors and earthing switches.

NF EN 62271-200 High voltage switchgear and control gear – A.C metal enclosed switchgear and control gear for rated voltage above 1kV and up to including 52kV.

NF EN 62271-103 Switches for rated voltage above 1kV up to and including 52kV.

NF EN 60529 Degree of protection provided by enclosures (IP code).

NF EN 61869-1 Instrument transformers – Part 1: General requirements.

NF EN 60044-1 Instrument transformers - Part 1: Current transformers.

NF EN 60044-2 Instrument transformers – Part 2: Voltage transformers.

NF EN 60815 Guide for selection of insulators in respect of polluted conditions.

NF EN 61000-6-4 Electromagnetic compatibility (EMC) – Part 6-4: Generic standards – Emission standard for industrial environments.

NF EN 61000-6-5 Electromagnetic compatibility (EMC) – Part 6-5: Generic standards - Immunity for equipment used in power station and substation environment.

NF EN 61869-2 Instrument transformers – Part 2: Additional requirements for current transformer.

NF EN 60947-6-2 Low-voltage switchgear and control gear – Multiple function equipment – control and protective switching devices.

ISO 4628-3 Paints and varnishes - Evaluation of deration of coatings - Designation of quantity and size of defects, and of intensity of uniform changes in appearance -- Part 3: Assessment of degree of rusting.

ISO 9001 (2007) Quality management systems – Requirements.

IEC 60099-4 Surge arresters - Part 4: Metal-oxide surge arresters without gaps for a.c. systems

DTP 562-3 Medium voltage installation and equipment – HTA cables.

1.3 Reference documents

[1]	B.2.2.a.v.5i	French Control System Specific Requirements (Rev 4.2 HIT-VIN Contract)
[2]	B.2.2.a.vi.1	Auxiliary Supply (France) (Rev 4.3)
[3]	1JNL1057068	Documentation Requirements For Biscay Gulf Project
[4]	1JNL2567050	Medium Voltage Switchgear

2 QUANTITIES AND ITEM DESIGNATION

2.1 France | Cubnezais | converter station 2

Five (5) switchgears, each consisting of cubicles which are listed below. The item designation of each cubicle shall be =S2.X1.Q1 for the first MVS and = S2.X2.Q1 for the second MVS.

S2.X1.Q1 (MVS-1)

= S2.X1.Q1.UC1 Incomer

= S2.X1.Q1.UC2 Measurement

= S2.X1.Q1.UC3 Outgoing feeder earthing transformer

= S2.X1.Q1.UC4 Outgoing feeder auxilary transformer TA1

= S2.X1.Q1.UC5 Coupler (from MVS-2)

S2.X2.Q1 (MVS-2)

= S2.X2.Q1.UC1 Incomer

= S2.X2.Q1.UC2 Measurement

= S2.X2.Q1.UC3 Outgoing feeder earthing transformer

= S2.X2.Q1.UC4 Outgoing feeder auxilary transformer TA1

= S2.X2.Q1.UC5 Coupler (from MVS-1)

S2.X9.Q1 (Enedis Grid DSO)

= S2.X9.Q1.**1 Incomer AI

= S2.X9.Q1.**2 Incomer AI

= S2.X9.Q1.**3 Voltage measurement TT

= S2.X9.Q1.**4 Coupling double switch breaker DDB

= S2.X2.Q1.UC1 Measurement

= S2.X2.Q1.UC2 Outgoing feeder auxilary transformer TA2 (terminal 1)

= S2.X2.Q1.UC3 Outgoing feeder auxilary transformer TA2 (terminal 2)

3 TECHNICAL DATAS

3.1 French RTE supplier (compliant to NF C 13-200)

3.1.1 Switchgear general

Description	Value
Installation	Indoor
Type of service	Continuous
Altitude	<1000 m
Temperature	-5°C / +42°C
Max. horizontal/Vertical acceleration	≤0,25/≤0,18 g.
Cubicle type	Primary or Secondary distribution
Setting	Single bar
Insulation medium	Air Insulated
Automatic Breaker	Vacuum circuit-breaker
Rated voltage	24 kV
Rated operating voltage	20 kV
Rated frequency	50 Hz, ± 5%
Number of phases	3
Rated insulation level	50 kV
Nominal insulation level kV impulse	125 kV
Rated normal current	630 A
Material in bus bar	Copper
Rated short-time withstand current	20 kA, 1 sec
Rated peak current	50 kA
Internal arc classification	IAC A FLR
Auxiliary voltage supply	125 Vdc
Protection degree	≥ IP3X
Color	RAL 7035
Phase indication	L1=red, L2=yellow, L3=blue

Phase sequence indication shall be marked all over the busbar conductors and on each cable terminations.

The switching devices shall be padlockable in closed position and in open position.

The disconnectors and the associated earthing switch shall be mechanically and electrically interlocked.

3.1.2 Circuits Breakers

Description	Value
Type	Not Withdrawable
Operating sequence of breaker shall be	O – 0.3s – CO – 3 min – CO
Rated voltage	24 kV
Rated operating voltage	20 kV
Rated insulation level	50 kVrms
Rated normal current	630 A
Rated short-circuit breaking current	20 kA
Rated short-circuit making current	50 kA
Type of breaker	Vacuum
Supply voltage closing/opening coil	125 Vdc
Supply voltage spring charging motor	125 Vdc
Auxiliary contacts: Open/Close	At least 4 NO and 4 NC

3.1.3 Disconnector/Earthing switches

Description	Value
Type	Manual operation
Rated voltage	24 kV
Rated operating voltage	20 kV
Rated insulation level	50 kVrms
Rated short-circuit breaking current	20 kA
Rated short-circuit making current	50 kA peak
Auxiliary contacts: Open/Close	At least 4 NO and 4 NC

3.1.4 Surge arrester

Description	Value
Type	Metal Oxide
Rated voltage	24 kV
Rated operating voltage	20 kV
Frequency	50 Hz
Line discharge class	Class 2
Shall comply with	IEC 60099-4

3.2 French Enedis supplier (compliant to NF C 13-100)

The switchboard shall be compliant with the standard NF C 13-100 with the Enedis special agreement HN 64-S-52 and HN 64-S-43.

3.2.1 Switchgear general

Description	Value
Installation	Indoor
Type of service	Continuous
Altitude	<1000 m
Temperature	-5°C / +42°C
Rated voltage	24 kV
Rated operating voltage	20 kV
Rated frequency	50 Hz, ± 5%
Number of phases	3
Rated insulation level	50 kV
Nominal insulation level kV impulse	125 kV
Rated normal current	400 A
Material in bus bar	Copper
Rated short-time withstand current	12.5 kA, 1 sec
Rated peak current	50 kA
Internal arc classification	IAC A FLR
Auxiliary voltage supply	48 Vdc
Protection degree	≥ IP3X
Color	RAL 7035

3.2.2 Functional Unit AI incomer

- Type 3 motorization for double derivation network according to specification HN 64-S-43.
- Connectable to a PASA/ITI cabinet - Auxiliary voltage 48 VDC included.
- Voltage presence relay type VDRI/U - Auxiliary voltage 48 VDC identical to the electrical control.
- 10 m long cord, ending with a male HA10 connector for connection to the PASA/ITI remote control cabinet.
- 2NO+2NC for switch position
- 2NO+2NC for earth disconnector position

3.2.3 Functional Unit TT Voltage measurement

- 3 VT 20kV/V3 - 2 x 100/V3 : 2 x 15 VA cl 0,5.
- 2NO+2NC for switch position
- 2NO+2NC for earth disconnector position
- 1 Decoupling protection relay C15-400 type ABB REU615

- 1 ESSAILEC voltage test box
- 1 RSE 3-position selector

3.2.4 Functional Unit DDB Coupling double switch breaker DDB

- A permanent three-phase voltage presence indicator block.
- 1 LV cabinet (left) with a four-pole fuse switch disconnector for the isolation and protection of metering LV circuits. Sealable by the energy distributor.
- 1 LV cabinet (right) with a four-pole fuse switch disconnector for the isolation and protection of metering LV circuits, for customer access.
- 2NO+2NC for switch position
- 2NO+2NC for earth disconnector position
- 2NO+2NC for circuit breaker position
- 3 CT 100-200 / 5-1
- 1 Decoupling protection relay C15-400 type ABB REF615
- 1 ESSAILEC voltage test box
- 2 ESSAILEC current test box

4 DESIGN

4.1 Switchgear assembly

The design of the switchgear shall be according to DTP 562-3.

The switchgear shall be of air insulated type.

The switchgear shall be factory assembled for indoor use.

The front door shall be safe for personnel at internal fault and withstand such as fault with damage and danger to person or property.

The apparatus shall be designed and tested to fulfil requirements of the applicable IEC standard. Deviations, if any, shall be indicated.

For all switchgears, no rotating gear shall be directly accessible when the local cabinet is opened. A plastic glass shall be added to protect operators from an unexpected operation of the mechanism. No key shall be necessary inside the local cabinet.

Entering of cables shall be possible from the bottom of the switchgear. The cubicles shall be provided with bottom seal plates.

Local open/close orders of circuit breakers shall be performed from the respective relay protection. One (1) Local/Remote switch shall be supplied for the complete switchgear assembly.

Open/close orders of circuit breakers shall normally be performed from remote location; circuit shall be wired to terminals and be available for the user for external connection.

Circuits for closing and opening/tripping the circuit breaker shall be separated. Power supply for the protections can be the same as the one for tripping the circuit breaker. There will be four (4) separate 125V DC for auxiliary supply. Suitable arrangement shall be made to connect these supplies on switch disconnectors ABB SD202/16 that shall be located next to the terminal blocks and included in the supply.

- C: Closing and indication
- O: Protection relay and tripping
- RA: Blocking
- M: Motor supply

All auxiliary contacts shall be connected to the terminals which are available for the user. The contacts shall be potential free.

All apparatus shall be marked with their position. The labels shall be easy to read and find. The item designation of each apparatus shall be documented in the list of apparatus. An earthing bar extending the whole length of the switchgear shall be provided.

Internal Arc Classification (IAC), A FLR – with ventilation down to the cable trench.

A Capacitive voltage indicator in all cubicles shall be included.

TF-E-VA and CC-E-VA test switches in all cubicles for current and voltage measurement shall be included.

Surge arrester shall be included in the delivery according to clause 4.12.

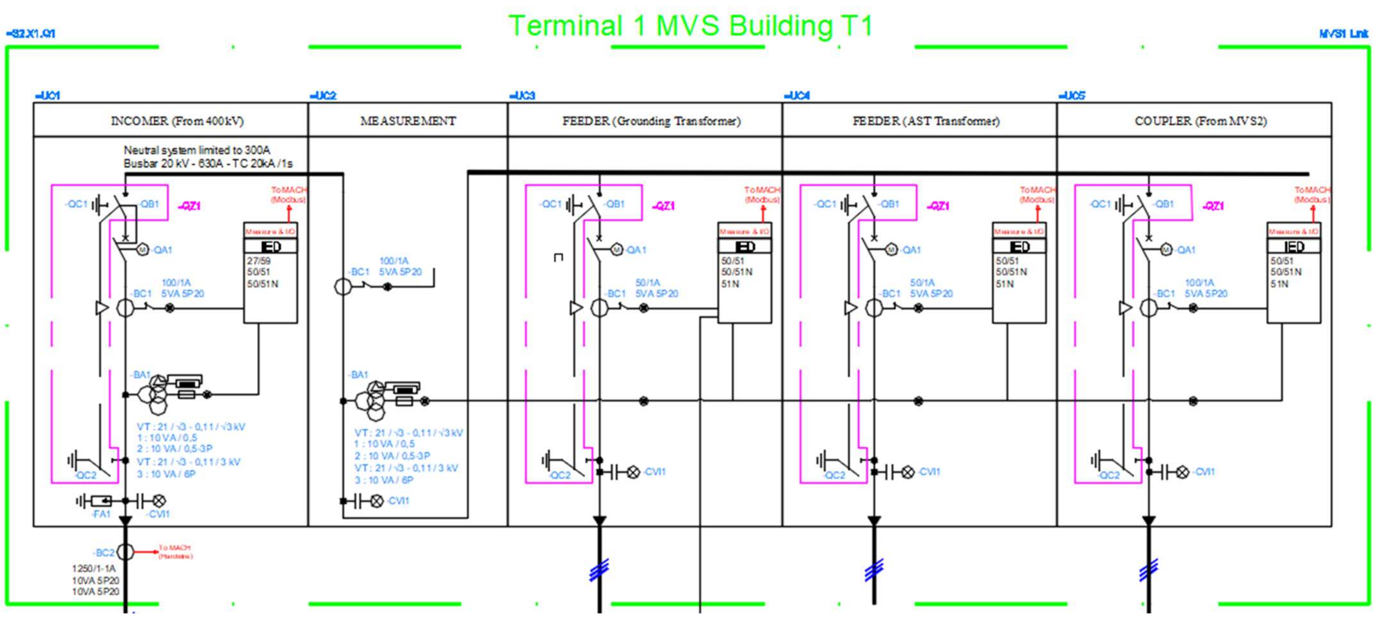
Phase sequence indication shall be marked all over the busbar conductors and on each cable terminations.

The switching devices shall be padlockable in closed position and in open position.

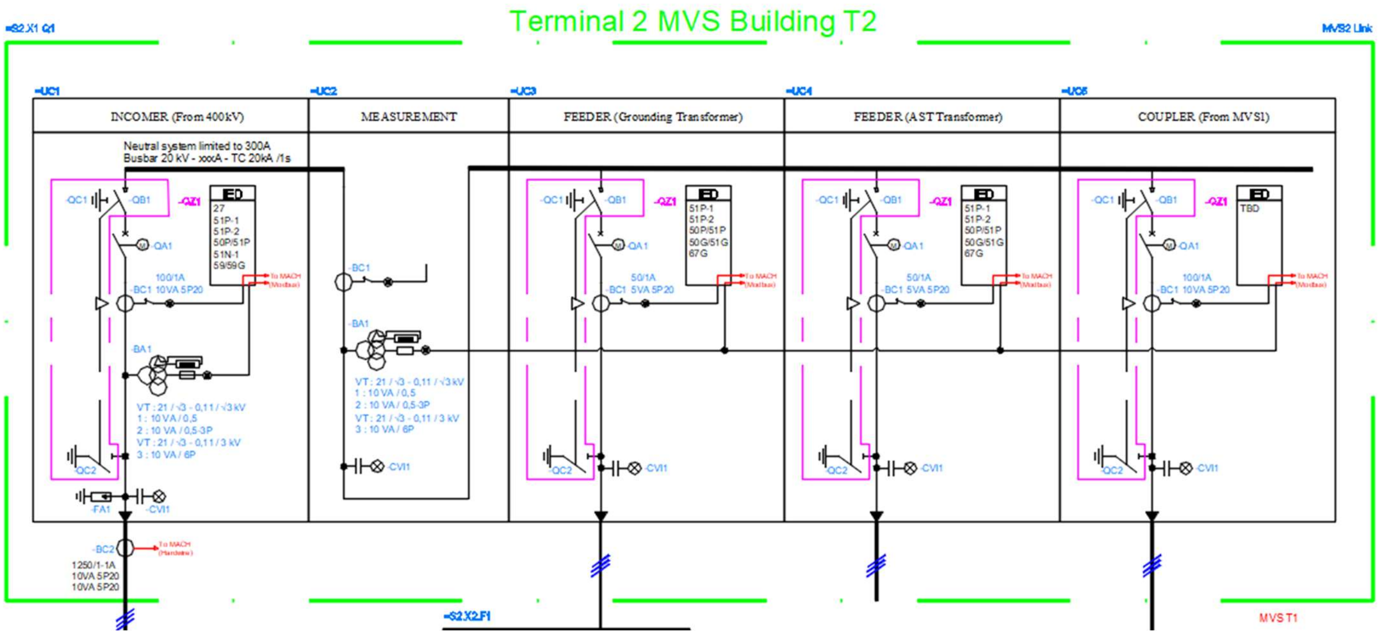
The disconnectors and the associated earthing switch shall be mechanically and electrically interlocked

4.2 Switchgear cubicle arrangement

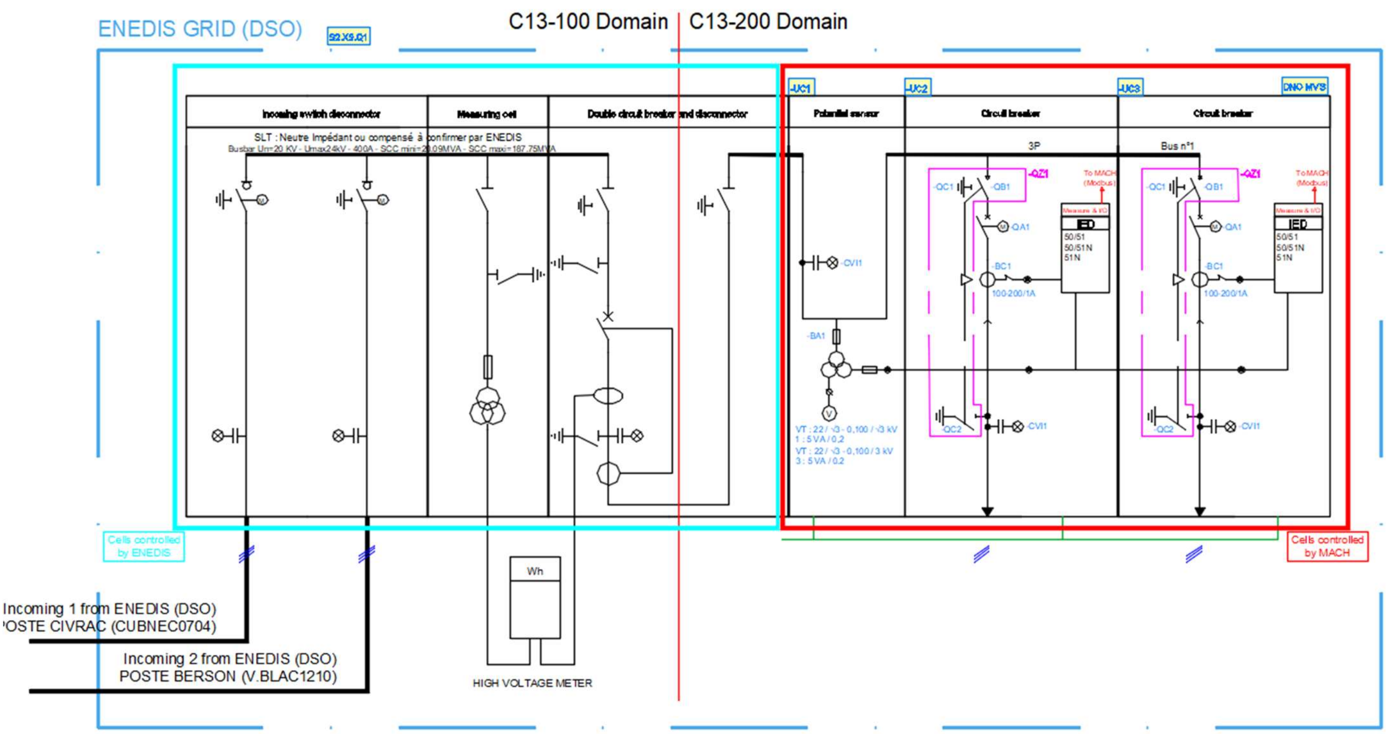
Bellow draft arrangement of MVS1



Bellow draft arrangement of MVS2



Bellow draft arrangement of Enedis grid



4.3 Circuit breakers

The circuit breakers shall be designed in accordance with NF EN 62271-100.

The mechanical endurance for the circuit breakers shall be at least M2 as per IEC 62271-100 and shall be chosen in accordance with their duty for a expected lifetime of 40 years.

The electrical endurance for the circuit breakers shall be at least E2 as per IEC 62271-100 and shall be chosen in accordance with their duty for a expected lifetime of 40 years.

The circuit breaker closing mechanism shall be of a motor spring charged operated type so that the closing speed is independent of the operator.

The spring-operated mechanism/circuit breaker shall have the following features:

- The operating sequence of breaker shall be O – 0.3s – CO – 3 min – CO.
- If the circuit breaker is closed with the springs charged and loss of auxiliary voltage for the operating mechanism occurs, the circuit breaker can be tripped, closed and tripped again.
- It shall be possible to charge the springs by hand. Where applicable there shall be visible indication of spring charged / un-charged.
- It shall not be possible to close the circuit breaker while the springs are being charged.
- It shall be necessary for the spring to be fully charged and the associated charging mechanism fully prepared for closing, before it can be released to close the circuit breaker.
- Tripping shall be affected by means of a trip coils.
- It shall be possible to trip the circuit breaker with two separate external trip signals, external trip should have a separate trip coil.
- The breaker position shall be indicated mechanically on breaker poles or operating device. The indication shall be legible during operation. The indication colour coding shall be according to IEC: green for open, red for closed. Position indicators shall be provided to indicate the open and close positions.
- Under normal conditions, motor recharging of the closing springs shall start immediately and automatically upon completion of each closing operation.
- Closing and tripping shall be performed by means of operating coils. However, it shall be possible to operate the breakers by push buttons on the cubicle front.
- The circuit breaker shall be equipped with necessary auxiliary contacts to obtain the required interlocking.

At least four (4) opening and four (4) closing true auxiliary contacts shall be available from circuit breaker and will be used for remote monitoring, control system, and electrical interlocking. The circuit breaker shall be equipped with an anti-pumping relay.

The circuit breaker shall be equipped with a counter for opening orders.

4.4 Disconnector / Earthing switches

Disconnector/earthing switches shall be designed in accordance with NF EN 62271- 102. Provisions for locking shall be included.

Grounding switches in all cubicles shall have an electrical interlocking. Also the disconnector on-load switch in both MVS-1 and MVS-2 incoming cubicle shall also have a provision of key interlocking system with upstream feeder from customer end.

If the incoming feeder is energized or if the MCB for the voltage transformer has tripped, it shall not be possible to close the grounding switch.

The disconnector and grounding switch closing mechanisms shall be manually operated.

Blocking solenoids shall be provided for the three-position switch in bays without a dedicated circuit breaker.

At least four (4) opening and for (4) closing auxiliary contacts shall be available from all disconnector/earthing switches and will be used for remote monitoring, control of system and electrical interlocking.

4.5 Mechanical interlocks and electrical interlocks

Circuit breaker and disconnector/earthing switch in same cubicle shall be mechanically interlocked. To prevent the removal of the cover of the cable compartment while the cable is not grounded, an interlocking solenoid shall be installed.

It shall be possible to electrically interlock the circuit breaker and earth switch in MVS with external equipment, i.e. earth switch and circuit breaker in LVS. Internal wiring and contacts shall be provided accordingly.

Link to interlocking scheme

4.6 Wiring Principles

The organization of the wiring shall be done as follows:

- Short circuit devices for CT secondary circuits where CT secondary circuits are used,
- VT secondary circuit opening device where VT secondary circuits are used,
- “Pre-wired system” circuits for VT secondary circuits, CT secondary circuits, closing/trip circuit for breakers. The term “pre-wired system” refers to a wiring bus that connects baseplates to which each equipment is connected with terminals equipped on ESSAILEC test plugs. For this specification, the term “pre-wired system” will be used,
- Terminals to external equipment (HV material, auxiliary systems…),
- A front mounted plate that is necessary for housing commutators for the maintenance of the cubicle see Figure 1:
    - One switch, called ICT, will be used to shut down all cubicle supplies and polarities including protection devices, control IEDs, other devices and wetting voltages with the exception for LAN switches power supplies. The ICT shall be in compliance with DTP S719. Reference shall be Kraus & Naimer CH10B FR1051*S6003 EF (see datasheet in Attachment).
    - o LAN switches supply will be only manageable from their own circuit breakers. Their circuit breakers will have to be placed in the same cubicles as they are,

- Internal wiring.

For the power supply of equipments, a pre-wired system can be used but it is not mandatory. In any cases, power supply must be easily shut down for each equipment and each equipment must use plugging devices that allow fast replacement.

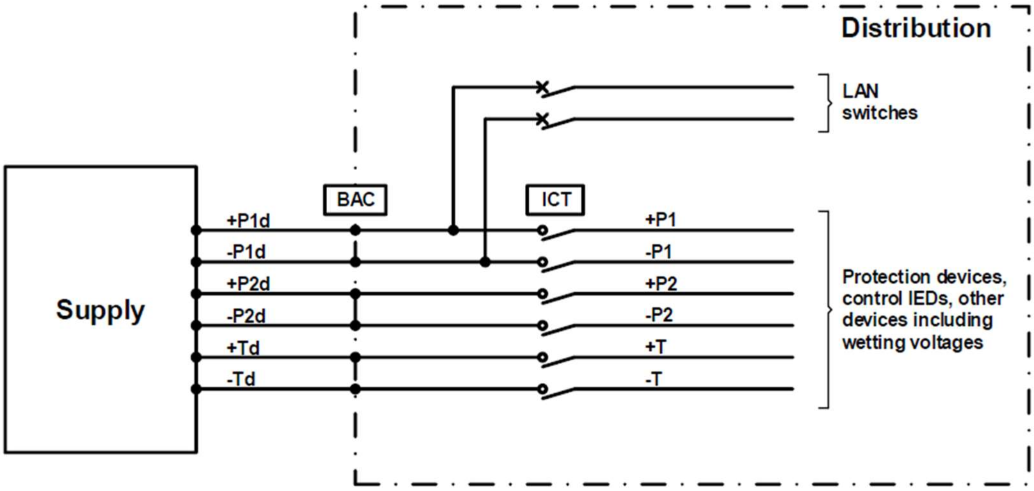


Figure 1: Internal power supply distribution

Labelling shall include the following nomenclature for phase numbering: N 0 4 8 or N 11 3 7 (as applicable), except for auxiliary power supply where phase numbering shall be N A B C.

Phase wiring order on terminal blocks, baseplates, equipment… shall follow the above order.

4.7 Measuring transformer

Cubicle	CT Value (TBD by calculation note)
S2.X1.Q1.UC1 S2.X2.Q1.UC1	100/1A 5VA 5P20 / cl 0.5
S2.X1.Q1.UC2 S2.X2.Q1.UC2	100/1A 5VA 5P20 / cl 0.5
S2.X1.Q1.UC3 S2.X2.Q1.UC3	50/1A 5VA 5P20 / cl 0.5
S2.X1.Q1.UC4 S2.X2.Q1.UC4	50/1A 5VA 5P20 / cl 0.5
S2.X1.Q1.UC5 S2.X2.Q1.UC5	100/1A 5VA 5P20 / cl 0.5
S2.X9.Q1.*** S2.X9.Q1.***	50/1A 5VA 5P20 / cl 0.5

Cubicle	VT Value (TBD by calculation note)
S2.X1.Q1.UC1 S2.X2.Q1.UC1	VT: 20 / √3 - 0,11 / √3 kV - 1: 10 VA / 0,5 - 2: 10 VA / 0,5-3P VT: 21 / √3 - 0,11 / 3 kV - 3: 10 VA / 6P
S2.X1.Q1.UC2 S2.X2.Q1.UC2	VT: 20 / √3 - 0,11 / √3 kV - 1: 10 VA / 0,5 - 2: 10 VA / 0,5-3P VT: 20 / √3 - 0,11 / 3 kV - 3: 10 VA / 6P
S2.X9.Q1.UC1	VT: 20 / √3 - 0,11 / √3 kV - 1: 10 VA / 0,5 - 2: 10 VA / 0,5-3P VT: 20 / √3 - 0,11 / 3 kV - 3: 10 VA / 6P

4.7.1 CT secondary circuits

The complete circuit from CT secondary circuit is described as follows:

- Secondary circuit from CT is connected in the cubicle on dedicated terminals. The connexion is done via 3xS1/S2 or 3xS1/1xS2-shunted and grounded cables. If the connexion is done with 3xS1/S2 cables, then the neutral is grouped in the cubicle. The internal distribution is done only with 4 wire circuits.
- The circuit goes to a short circuit device. This device gives an information when the circuit is short circuited. This information is integrated into the C&P system.
- The pre-wired system follows with the following:
    - A testing baseplate: this is a baseplate that when in use with the test plug, does the short circuit on the CT side and allows injection from external source on the other side. (PE-I)
    - Baseplates for each equipment (IED, protection...) using CT circuit. (I-IED)
    - Terminal baseplate: after this baseplate must be realized the final short-circuiting of the current circuit. (I-TERM)
    - o Each IED or protection is then connected to the circuit using ESSAILEC plugs connected to relative baseplate.

When not in use, all baseplates shall be fitted with protective cover.

Those circuits are distributed by 4 conductors (3Ph+N). The section of conductor on the baseplate circuits shall be as a minimum 4mm². The section on the plug to the equipment shall be 2.5mm². The connection to baseplate shall follow the order described in DTP S702: N 0 4 8 or N 11 3 7 (as applicable).

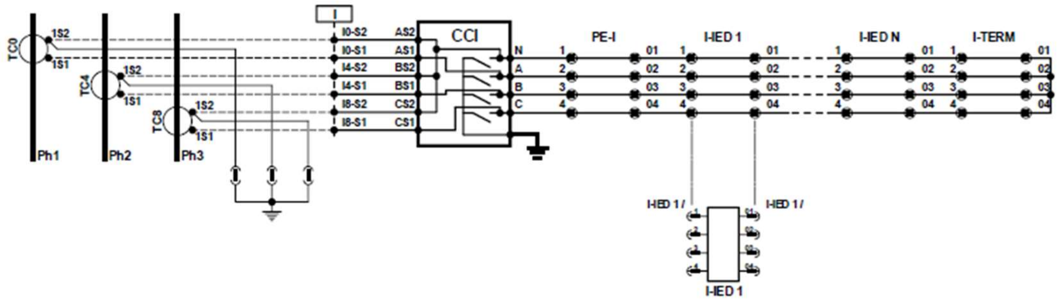


Figure 2: Principle for the pre-wired system and IED wiring for CT secondary circuit

Material

Material	Reference	Employer Specification
Short circuit device	SOCOMEC 2935 0001	DTP S721 See datasheet in Attachment
Testing baseplate	ENTRELEC CC-E-VA 1SNA166737R2000 Green baseplate Built-in system Internal circuit shunted	DTP S702 See catalogue in Attachment
Equipment baseplates	ENTRELEC CC-D-VA 1SNA166738R0100 Green baseplate Semi built-in system Internal circuit shunted	DTP S702 See catalogue in Attachment
Terminal baseplate	ENTRELEC CC-D-VA 1SNA166738R0100 Green baseplate Semi built-in system Internal circuit shunted	DTP S702 See catalogue in Attachment
Plug tests	ENTRELEC FIC-2/4-2 1SNA166936R1000 Green plug test	DTP S702 See catalogue in Attachment

4.7.2 VT secondary circuits

The complete circuit from VT secondary circuit is described as follows:

- Secondary circuit from VT is connected in the cubicle on dedicated terminals. The connexion is done via 3xS1/S2 or 3xS1/1xS2-shunted and grounded cables. If the connexion is done with 3xS1/S2 cables, then the neutral is grouped in the cubicle. The internal distribution is done only with 4 wire circuits.
- The circuit goes to an circuit opening device,
- The pre-wired system follows with the following: o A testing baseplate, (PE-U)
  - Baseplates for each equipment (IED, protection..) using VT circuit, (U-IED)
  - Terminal baseplate: after this baseplate, the VT circuit must be opened, (U-TERM)
  - Each IED or protection is then connected to the circuit using ESSAILEC plugs connected to relative baseplate.

When not in use, all baseplates shall be fitted with protective cover.

The VT circuit opening device shall allow protection and opening of the secondary circuit for VT. The opening of the circuit shall be visible and lockable in open position. It will be realized by means of a fuse holder which conception shall allow the replacement of a fuse without opening all the phases.

The protection of voltage circuits shall answer the following criteria:

- Permanent current supported : 0.2 to 1A,

- Breaking capacity : 280A,
- Fault Elimination time: < 250ms pour 280A,
- Stability: in case of oscillations that can attend 25A peak in 200ms, the dispositive shall not trip.

Fuses shall be individually monitored.

Fuses holder shall allow to open the neutral.

Fuses holders shall be SOCOMEC RMSC 57C2 5034 (3P+N model) or SOCOMEC RMSC 57C2 5032 (P+N model).

Incoming voltage signal cables shall be connected to fuses holder from the top.

Connection of fuse break monitoring signal cable shall be IP2X.

Those circuits are distributed by 4 conductors (3Ph+N). The section of conductor on the baseplate circuits shall be as a minimum 4mm². The section on the plug to the equipment shall be 2.5mm². The connection to baseplate shall follow the order described in DTP S702: N 0 4 8 or N 11 3 7 (as applicable).

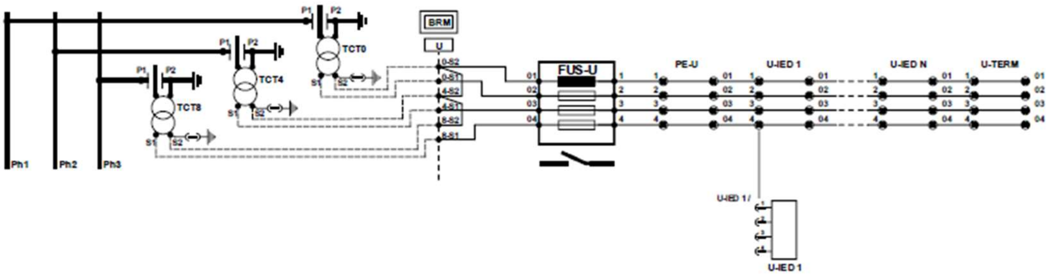


Figure 3: Principle for the pre-wired system and IED wiring for VT secondary circuit

Material

Material	Reference	Employer Specification
VT opening device	SOCOMEC RMSC 57C2 5034	See below and datasheet in Attachment
Testing baseplate	ENTRELEC TC-E-VA 1SNA166747R0200 Grey baseplate Built-in system Internal circuit shunted	DTP S702 See catalogue in Attachment
Equipment baseplates	ENTRELEC TF-D-VA 1SNA166746R0100 Grey baseplate Semi built-in system Internal circuit closed	DTP S702 See catalogue in Attachment
Terminal baseplate	ENTRELEC TF-D-VA 1SNA166746R0100 Grey baseplate Semi built-in system Internal circuit closed	DTP S702 See catalogue in Attachment
Plug tests	ENTRELEC FIT-2/4-2 1SNA166937R1100 Gray plug test	DTP S702 See catalogue in Attachment

4.8 Protection Relay

The protection relay type shall be ABB REX615 numerical type with Modbus TCP communication module and not trip in case of an internal fault or communication error. Single Line Diagram shall be shown on IED's HMI.

All IED signals shall be connected through test switch. The software for the relays and cable for loading the settings shall be part of the delivery.

The cubicles shall be provided with the following relay functions. The functions shall be integrated in one relay unit. All available contacts and functions shall be wired to terminals and be available for the user. Auxiliary contacts shall be potential free.

Lien Plan de protection

MVS LINK 1

Incomer power transformer tertiary winding 400/20kV	27/59: Undervoltage Relay / Overvoltage Relay 50/51: Phase Instantaneous Overcurrent / Phase Time Overcurrent 50/51N: Neutral Instantaneous Overcurrent / Neutral Time Overcurrent 86: Locking-Out Relay 25: Synchro check ???
Feeder 1: Earthing transformer	50/51: Phase Instantaneous Overcurrent / Phase Time Overcurrent 50/51N: Neutral Instantaneous Overcurrent / Neutral Time Overcurrent 50/51G : Ground Instantaneous Overcurrent / Ground Time Overcurrent (Grounding transformer) 86: Locking-Out Relay
Feeder 2: Auxiliary transformer AST 1	50/51: Phase Instantaneous Overcurrent / Phase Time Overcurrent 50/51N: Neutral Instantaneous Overcurrent / Neutral Time Overcurrent 51N : Neutral Time Overcurrent (Grounding transformer) 86: Locking-Out Relay
Coupling from MVS2	27/59: Undervoltage Relay / Overvoltage Relay 50/51: Phase Instantaneous Overcurrent / Phase Time Overcurrent 50/51N: Neutral Instantaneous Overcurrent / Neutral Time Overcurrent 86: Locking-Out Relay 25: Synchro check ???

MVS LINK 2

Incomer power transformer tertiary winding 400/20kV	27/59: Undervoltage Relay / Overvoltage Relay 50/51: Phase Instantaneous Overcurrent / Phase Time Overcurrent 50/51N: Neutral Instantaneous Overcurrent / Neutral Time Overcurrent

	86: Locking-Out Relay 25: Synchro check ???
Feeder 1: Earthing transformer	50/51: Phase Instantaneous Overcurrent / Phase Time Overcurrent 50/51N: Neutral Instantaneous Overcurrent / Neutral Time Overcurrent 50/51G : Ground Instantaneous Overcurrent / Ground Time Overcurrent (Grounding transformer) 86: Locking-Out Relay
Feeder 2: Auxiliary transformer AST 1	50/51: Phase Instantaneous Overcurrent / Phase Time Overcurrent 50/51N: Neutral Instantaneous Overcurrent / Neutral Time Overcurrent 51N : Neutral Time Overcurrent (Grounding transformer) 86: Locking-Out Relay
Coupling from MVS1	27/59: Undervoltage Relay / Overvoltage Relay 50/51: Phase Instantaneous Overcurrent / Phase Time Overcurrent 50/51N: Neutral Instantaneous Overcurrent / Neutral Time Overcurrent 86: Locking-Out Relay 25: Synchro check ???

ENEDIS LINK (DSO)

TBD	TBD
Feeder 2: Auxiliary transformer AST 1	50/51: Phase Instantaneous Overcurrent / Phase Time Overcurrent 50/51N: Neutral Instantaneous Overcurrent / Neutral Time Overcurrent 51N : Neutral Time Overcurrent (Grounding transformer) 86: Locking-Out Relay
Feeder 2: Auxiliary transformer AST 1	50/51: Phase Instantaneous Overcurrent / Phase Time Overcurrent 50/51N: Neutral Instantaneous Overcurrent / Neutral Time Overcurrent 51N : Neutral Time Overcurrent (Grounding transformer) 86: Locking-Out Relay

4.9 Low voltage cabinets

Description	Value
Lighting system	LED
16A power socket	No (socket available in the building – SPL)
Local/Remote	Selector switch
Heating system	Yes

4.10 Instruments

Each MV cubicles shall be able to measure active and reactive power either through protection and control IED or through a measurement system. These measurements shall be integrated within C&P system in order to be accessible locally through HMI system or remotely through remote access.

4.11 Bus communication & signal interface

A communication module shall be included in IED. Fibre optical interface with Modbus TCP. It should be possible to connect 2 FO patch cable (A & B) from external communication cabinet. Measuring signals and indications should be sent over MODBUS and will be extended to MACH system.

Lien Signal List

4.12 Miniature circuit breaker (MCB's)

MCB's shall be supplied for the secondary circuits of the voltage transformers and for the auxiliary circuits, for the control circuit and motors. The MCB's shall be provided with auxiliary contacts and padlocking systems. MCB Tripped/Off signal shall be a common signal for all MCB's.

The current rating of the devices shall be chosen according to the load they protect in order to prevent selectivity issues upstream.

A functional description of each circuit shall be indicated on or near its MCB for clarification in a durable manner. Approval by Client required.

4.13 Surge arresters

Surge arrester shall be of Plug- in type, metal enclosed, easy for maintenance and replacement and shall comply with IEC 60099-4.

The enclosure is reliably earthed and safe for touching.

4.14 Terminal blocks

Terminal blocks shall be selected based on the following table:

Type	Supplier	Use	Applications
OTTA6	Phoenix	Current measuring	CT Junction box, phase current, earthing
UK10N	Phoenix	Power supply and Voltage measuring	Power supply, CVT interface
M4/6.ST.SN M10/10.ST.SN	TE	Digital Input/Output	Circuit breaker open order, alarm interface, indication interface, order interface, 4-20 mA, PT100

For applications not listed in the above table, Contractor shall select terminal block reference from this table which suits the most the application. If application require different terminal blocks than listed in this table, Contractor shall propose a reference or a set of references for Employer review and approval.

When insertion bridge is used, it shall not be inserted for a continuous duration longer than 24 hours. This requirement shall apply during all phases, including storage and transport phases. Should this requirement not be met, the Employer will request the Contractor to replace the terminal block at Contractor's own expenses.

Multi-level terminal blocks shall not be used.

4.15 Protection co-ordination

In order to obtain full information for the selectivity plans; the Supplier shall provide all required information about the MV switchgear and protection relays to ABB AB.

4.16 Degree of protection

The switchgear shall be designed so that the degree of protection, according to NF EN 60529, corresponds to IP31 minimum.

4.17 Seismic requirement

According to 1JNL552308 [2] - General requirements for Auxiliary power system.

4.18 Single line diagram for Cubnezais Station

According to 1JNL652623 [1] – Single Line Diagram station overview for Cubnezais station.

4.19 Surface treatment

The surface treatment shall be able to withstand at least environment class C1 as per 1JNL552308 [2] - General requirements for Auxiliary power system.

The colour shall be light grey RAL7035.

5 NAMEPLATES

6 TESTS

6.1 ROUTINE TESTS

The routine tests shall be performed according to NF EN 62271.

6.2 TYPE TESTS

The type tests shall be performed according to NF EN 62271, the constructor can supply the existing type test summary report for these equipment.

7 MANUALS AND DOCUMENTATION

7.1 Technical documentation

Within the first month from the date of the order, the Supplier will submit to client's approval the following plans, along with their list (in accordance with the regulations):

- Descriptive report of the equipment design and manufacturing procedure.
- List of exceptions to this specification, including appropriate justifications or considerations. Exceptions not included will not have contractual validity.
- Indicative general dimension plan with arrangement of elements.
- Electrical functional schemes and complete wiring diagrams.
- When the use of equivalent previous designs is proposed, the list of the most relevant type test certificates available will be indicated in the offer.
- Additional information related to components and accessories susceptible to use not specifically included in this specification.
- Appraised list of special tools necessary for assembly and maintenance.
- Rated list of recommended spare parts for 5 years.
- Approximate manufacturing and shipping schedule.

According to documentation requirements for Biscay Gulf Project [5].

All documentation must indicate the equipment serial number as well as the order number.

All documents will be delivered in English and French, and once approved, the supplier will deliver the specific documentation files in .DWG (AutoCAD) and Microsoft Office format.

7.2 Operation Manual

The operational manual should be provided for each auxiliary system and should contain the system’s description (with each equipment), relevant information on how each system works, has to be operated (start/stop sequence, parameter or status modification, …) and interacts with other systems.

These manuals should also contain screenshots with step by step explanation on where to click and how to proceed for each sequence on the dedicated HMI.

It will be necessary to provide the Replacement Sequence manual of all equipment and how it affects the operation of the System.

7.3 Commissioning and installation manual

There shall be a dedicated commissioning and installation manual for each auxiliary system equipment.

This manual should describe how to assemble the equipment, describe storage and transportation constraints as well as the pre-commissioning and commissioning tests, handling of the equipment in a safe manner, etc.

Configuration manuals shall be also part of the commissioning and installation manual: including all the documentation required for the configuration of each device of the control system which can be performed by qualified and trained crew of the Employer.

7.4 Preventive Maintenance Handbook

A preventive maintenance handbook for each equipment or family of equipment of the auxiliary power system of the Converter Station shall be provided.

The preventive maintenance handbooks shall accurately cover in detail each maintenance task listed in the Maintenance Activity List (MAL)/ Maintenance and Inspection Plan (MIP). Each maintenance task shall be classified by frequency (monthly, quarterly, yearly etc…). The aim of this part of the manual is to describe step by step by means of pictures of the actual on site equipment how to proceed for each maintenance task mentioned in the MAL/MIP.

7.5 Corrective Maintenance Handbook

A Instruction Manual for each equipment or family of equipment of the auxiliary power system of the Converter Station shall be provided. These instruction manuals shall cover all the corrective maintenance aspects related to their respective equipment, and especially include a detailed procedure for equipment replacement.

For the C&P (control and protection) equipment, additional documents, called "corrective maintenance handbooks" shall be provided. For the details of this manual, refer to B.2.2.c - Maintenance Specification.

8 SPECIAL TOOLS AND SPARE PARTS

The spare parts, special tools and maintenance criteria will follow the instructions indicated and developed

in B.2.2.c - Maintenance Specification.

A complete set of spares, including maintenance kit (special tools and test equipment), required for at least five years of operation to meet the Guaranteed Failure Rates, will be provided for each equipment defined in this technical specification.

The Spare Parts will have the same physical, mechanical and electrical properties as the installed Equipment on Site and shall be subject to the same tests as the Equipment on Site.

Any spares and maintenance kit used during the Warranty period during delivery of maintenance activities shall be replenished as part of the Warranty period at the manufacturer’s cost, such that the Spare Parts stock shall be fully restocked before the end of the Warranty period.

Any spares parts with a limited self-life, i.e. sealing compounds, lubrication and fluids, shall be within date at the end of the Warranty period.

The manufacturer will provide a list of spare parts including maintenance kids, that are considered necessary for the equipment provided. This spare list will be included in the Spare Parts List (SPL) of the Converter Station. Some equipment has already defined spare parts in the document: 1JNL1100559_en BGP Customer spare part.